MD. SAZZAD HOSSAIN

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RESEARCH

Power-system modeling and stability, renewable-energy integration, physics-informed and self-calibrating digital twins, real-time parameter estimation of inverter-based resources, machine learning for power-system monitoring and resilience, and intelligent grid control.

EDUCATION

American International University-Bangladesh — Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering — Jan 2022 – Dec 2025
Final grade: 3.57/4.00 | Credits: 148

PUBLICATIONS

Conference papers

1. Shaharier Kabir, **Md. Sazzad Hossain**, Md Sultan Mahamud, Tasnim Kabir, Abu Shufian, Muhammad Masud Karim, Saniat Rahman Zishan, "*A Physics-Informed Self-Calibrating Digital Twin with Conformal Uncertainty for Multi-Inverter PV Power Systems*," IEEE TENSYMP, Penang, Malaysia, 29 Jun – 1 Jul 2026 (Presented, 2nd author).
2. Asif Iqbal, Sumit Hassan Eshan, Md Farukuzzaman, **Md Sazzad Hossain**, Azizul H Tuhin, Sadia Ahmed, Samiul Hasan, Orpyta Karmokar Aupy, "*Optimized Transport Layers for Enhanced Efficiency in Perovskite-Perovskite Tandem Solar Cells*," PEEIACON, Southeast University (SEU), Dhaka, Bangladesh, 4 – 5 Dec 2025 (Presented, 4th author).

Accepted

1. **Md. Sazzad Hossain**, Apurba Joty Roy, Shaharier Kabir, Abu Shufian, "*Recursive Bayesian Self-Calibration and Conformal Uncertainty Bounds for Solar-PV Digital Twins Under Grid Variability*," IEEE PES ISGT Asia 2026, Wuhan, China, 30 Oct to 1 Nov 2026 (Accepted). https://attend.ieee.org/isgt-asia-2026/

Under review

1. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Md Aktar Hossain, Md. A. Hamid Howlader, "*A Stability-Aware Demand-Side Management (DSM) Framework for Grid-Forming Inverter Transient Stability Under EV Charging Disturbances*," Kollam, Kerala, India (Submitted). https://conference.ukfcet.ac.in/#home
2. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Raihan Chowdhury, Goro Fujita, "*Calibrated Staleness Detection for Solar Digital Twins and the Dominance of the Retraining Window over the Detection Trigger*," NSysS, Cox's Bazar, Bangladesh (Submitted). https://cse.buet.ac.bd/nsyss2026/
3. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Raihan Chowdhury, Goro Fujita, "*A Multi-Cell Physics-Based Digital Twin with Closed-Loop Kalman Filtering and Remaining-Useful-Life Prognosis for Lithium-Ion Battery State of Health*," NSysS, Cox's Bazar, Bangladesh (Submitted). https://cse.buet.ac.bd/nsyss2026/
4. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Md Aktar Hossain, Md. A. Hamid Howlader, "*Stability-Aware Distributed Stability Margin for Grid-Forming Inverter Small-Signal Stability*," Wuhan, China (Submitted). https://attend.ieee.org/isgt-asia-2026/

Submitted

1. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Raihan Chowdhury, Goro Fujita, "*Inverse Hot-Spot and Power Estimation in GaN HEMTs Using a Green's Function-Augmented Physics-Informed Neural Network*," Dhaka, Bangladesh (Submitted). https://icece.org.bd/2026/

JOURNAL

Under review

1. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Debashish Kumar Ghosh, Md. Sajid Hossain, "*Digital-Twin Staleness Detection in Renewable Generation: A False-Alarm-Matched Benchmark of Sequential Drift Detectors across Sixteen European Fleets*," Next Sustainability, manuscript NXSUST-D-26-01704 (Submitted).

In preparation

1. *A Conformalized Digital-Twin Framework for Trustworthy False Data Injection Attack Detection in Cyber-Physical*

Power Systems, Ongoing (In preparation).

PROJECTS

AI-Based Motion Detection Edge Computing System Apr 2024 – May 2025

- Designed an intelligent edge computing system that captures and processes images using computer vision algorithms to detect motion through sequential frame analysis.
- Implemented AI-based classification on captured images to identify motion events and trigger automated responses.
- Integrated an AI camera module with Raspberry Pi 5 and a vision kit for complete on-device processing without cloud dependency.
- Optimized the image-capture pipeline and motion-detection algorithms for low-latency performance on edge hardware.

WORK

Energypac Engineering Ltd. Dhaka, Bangladesh
Internship Trainee Jun 2025 – Jul 2025

- Gained hands-on experience in the manufacturing and testing of power equipment including transformers, switchgear, and protective devices.
- Assisted with equipment assembly, relay settings, and quality inspections.
- Participated in high-voltage testing procedures in accordance with IEC standards.
- Supported testing and commissioning activities for power distribution equipment.

SKILLS

Programming & Data Tools: C, C++, Python, NumPy, Pandas, SciPy, Matplotlib, Seaborn, TensorFlow, PyTorch, Scikit-learn, Git/GitHub, OpenCV, MATLAB & Simulink, Visual Studio Code, Google Colab

Documentation & Typesetting: LaTeX (Intermediate), Overleaf, Microsoft Office

Hardware Design, Circuits & PCB: EasyEDA, NI Multisim, Proteus

EDA, IC & CAD Tools: Cadence Virtuoso, AutoCAD, AutoCAD Electrical, SOLIDWORKS

Simulation, RF/EM & Optics: CST Microwave Studio, Antenna Design, OptiSystem

Graphic Design & Illustration: Inkscape, CorelDRAW

Energy Systems: Renewable Energy, HOMER Energy, RETScreen, PSCAD, PSS/E, DIgSILENT PowerFactory

Embedded Systems: Raspberry Pi 5, ESP32, Arduino Uno, AI Vision Kit, Motion Sensors

Professional Skills: Research Skills, Problem Solving, Team Leadership, Project Management, Presentations, Marketing

LANGUAGES

Mother tongue: Bengali
Other languages: English, Hindi, Urdu